

# SHIN, Young Ah

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## RESEARCH INTERESTS

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*"My research focuses on cryptographic approaches to enhancing trust and privacy in emerging technologies — particularly Internet-of-Things, federated learning, and blockchain — through the design of advanced cryptographic primitives and privacy-preserving protocols with provable security guarantees."*

### Key Research Areas:

- Digital Signatures (ID-Based Signature, Proxy Signature, Multi-Signature, Adaptor Signature, etc.)
- Lightweight Cryptographic Protocols for Mobile Devices
- Privacy-Preserving Federated Learning Framework
- Privacy-Preserving and Scalable Blockchain Payment Protocols

## EDUCATION

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### Ph.D.

#### Korea University (QS #61), School of Cybersecurity

Mar. 2020 — Aug. 2026 (TBC)

**Laboratory:** Privacy-Enhancing Technologies (PET) Lab.

**Advisor:** Prof. Ik Rae JEONG, Dean at School of Cybersecurity

**Dissertation:** Practical Secure Design of Online and Offline Cryptographic Protocols for Mobile Application

Designed practical, provably secure on/offline cryptographic protocols for mobile applications, focusing on identity-based multi-proxy signatures and on/offline signcryption in Internet-of-Drones (IoD) environments.

- Established the necessity of separating proxy signing keys from long-term private keys in public-key settings; proposed a provably secure identity-based multi-proxy signature scheme and extended it to on/offline protocols
- Identified a previously overlooked memory corruption threat between online and offline phases in multi-device environments; proposed two identity-based on/offline signcryption schemes with provable security against such attacks

### B.S.

#### Sungshin Women's University, Department of Convergence Security Engineering

Mar. 2016 — Feb. 2020

**Advisor:** Prof. Il-Gu LEE, Distinguished Professor at Sungshin Women's University

**Thesis:** Blockchain-Based Mobile Coupon Trading Platform for Campus and Retailers Using zk-SNARKs

## PUBLICATIONS

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### International

- [In Progress] **Young Ah Shin**, Ik Rae Jeong, "Bitcoin-Compatible and Anonymous Payment Channel Hub with Fine-Grained Auditing (TBD)," (To Be Submitted in Q3 2026)
- [In Progress] **Young Ah Shin**, Ik Rae Jeong, "Unpredictable Leader Election for Decentralized and Robust Multi-Party Payment Channel," (To Be Submitted in Q2 2026)
- [Under Review] **Young Ah Shin**, Ik Rae Jeong, Jin Wook Byun, "Secure ID-Based On/Offline Signcryption from Online and Offline Attacks," (Under Review at **IEEE Transactions on Dependable and Secure Computing** (Q1, 2024 IF: 7.5), submitted in Q1 2026)

- **[Under Review]** Joon Ho Lim, **Young Ah Shin**, Ik Rae Jeong, "PETRA: Privacy-preserving Efficient Tracing for Regulation-compliant CBDC," (Under Review at **IEEE Transactions on Information Forensics and Security (Q1, 2024 IF: 8.0)**, submitted in Q4 2025)
- **[Published]** Gyeong Min Baek, **Young Ah Shin**, Gye Hyun Jang, Ik Rae Jeong, "Block Number-Based Address Clustering for Bitcoin Taproot Upgrade," **IEEE Access (Q1, 2024 IF: 3.6)**, vol. 13, pp. 94523 – 94537, 2025.  
DOI: 10.1109/ACCESS.2025.3570459 ([link](#))
- **[Published]** **Young Ah Shin**, Ik Rae Jeong, Jin Wook Byun, "Identity-Based Multiproxy Signature with Proxy Signing Key for Internet of Drones," **IEEE Internet of Things (Q1, 2022 IF: 10.6)**, vol. 11, no. 3, pp. 4191-4205, 2023.  
DOI: 10.1109/JIOT.2023.3300299 ([link](#))
- **[Published]** **Young Ah Shin**, Geontae Noh, Ik Rae Jeong, Ji Young Chun, "Securing a Local Training Dataset Size in Federated Learning," **IEEE Access (Q1, 2021 IF: 3.476)**, vol.10, pp.104135-104143, 2022.  
DOI: 10.1109/ACCESS.2022.3210702 ([link](#))

## INTERNATIONAL COLLABORATION & RESEARCH ACTIVITIES

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### International Conferences & Engagement

**Korea University International Conference on Cybersecurity (KU-ICCSP 2025)** ([link](#)) *Oct. 27-28, 2025*

- Served as a Student Representative of Organizing Committee
- Coordinated logistics and communication with 24 speakers and session moderators across 8 countries
- Managed diplomatic outreach to 16 ambassadors and 24 diplomats in Seoul for Opening Ceremony attendance
- Led an 18-member student operations team responsible for segment-wise planning, and on-site execution

**EU Study Visit on Cybersecurity Skills & Higher Education** ([link](#))

*Mar. 12, 2025*

- Provided communication support and coordinated campus tours

### International Research Collaboration

**The APAC Cybersecurity Fund by The Asia Foundation & Google.org**

*Mar 2026 — Present*

- Participating in the APAC Cyber Clinic Consortium, including experience sharing on cyber clinic management
- Serving as a member of the Korea University Cyber Security Clinic Center
- Planning MSME outreach, recruiting 34 KU students, training, and managing outreach & consulting activities

**University Blockchain Research Initiative (UBRI) by Ripple**

*Jun. 2023 — May 2026*

- Awarded a grant and participated in the global XRP Ledger (XRPL) ecosystem community, including APEX 2025
- Operated a Ripple Validator node and managed the Blockchain Security Research Center at Korea University

## TEACHING EXPERIENCE

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**PDP008: Privacy by Design**

*Mar. 2025 – Present*

**Institution:** Korea University

**Department:** Graduate School of Privacy & Data Protection | **Language:** Korean | **Credits:** 2

Delivered lectures on Privacy by Design (PbD) principles and strategies, usable security and privacy, and case study analysis, emphasizing privacy, security, and usability considerations from the design stage to working professionals

**PDP003: Privacy Enhancing Technology**

*Mar. 2025 – Present*

**Institution:** Korea University

**Department:** Graduate School of Privacy & Data Protection | **Language:** Korean | **Credits:** 2

Delivered introductory topics in modern cryptography and privacy-enhancing technologies (PETs), including PKI, zero-knowledge proofs, blockchain, and federated learning to working professionals

## RESEARCH PROJECT EXPERIENCE

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>[Ongoing]</b> "APAC Cybersecurity Fund Cyber Clinics"<br><b>Funded by:</b> The Asia Foundation, supported by Google.org                                                                                                                                                                                              | <i>Mar. 2026 – Present</i>   |
| <b>[Ongoing]</b> "Development of Digital Asset Transaction Tracking Technology to Prevent Malicious Financial Activities in the Digital Asset Market"<br><b>Funded by:</b> Institute of Information & Communications Technology Planning & Evaluation (IITP) in collaboration with U.S. Department of Homeland Security | <i>Jul. 2024 – Dec. 2026</i> |
| <b>[Ongoing]</b> "University Blockchain Research Initiative (UBRI)"<br><b>Funded by:</b> Ripple                                                                                                                                                                                                                         | <i>Jun. 2023 – May 2026</i>  |
| <b>[Completed]</b> "Analysis and Improvement of CBDC Ensuring Both Auditability and User Privacy"<br><b>Funded by:</b> Korea Institute of Information Security & Cryptology (KIISC)                                                                                                                                     | <i>Feb. 2025 – Nov. 2025</i> |
| <b>[Completed]</b> "Study on the Security of Vector Encryption Schemes"<br><b>Funded by:</b> Viva Republica                                                                                                                                                                                                             | <i>Feb. 2025 – Aug. 2025</i> |
| <b>[Completed]</b> "Research, Analysis, and Consulting on the Establishment and Integrated Control of Anti-Drone Systems in Trade Ports"<br><b>Funded by:</b> Ministry of Oceans and Fisheries                                                                                                                          | <i>Apr. 2024 – Oct. 2024</i> |
| <b>[Completed]</b> "Analysis and Security Enhancement of Multi-Signature Schemes Used in Blockchain"<br><b>Funded by:</b> Korea Institute of Information Security & Cryptology (KIISC)                                                                                                                                  | <i>Feb. 2023 – Nov. 2023</i> |
| <b>[Completed]</b> "Analysis of Changes in the Non-Proliferation Regulatory Environment and Research on Policy and Technology Directions with the Convergence of 4th Industrial Revolution Technologies"<br><b>Funded by:</b> Korea Foundation of Nuclear Safety (KOFONS)                                               | <i>May 2022 – Oct. 2022</i>  |
| <b>[Completed]</b> "Design of a Blockchain-Based Cryptographic Module Certificate Management System"<br><b>Funded by:</b> National Security Research Institute (NSR)                                                                                                                                                    | <i>Apr. 2022 – Oct. 2022</i> |
| <b>[Completed]</b> "Military Cryptography Research Center"<br><b>Funded by:</b> Agency for Defense Development (ADD)                                                                                                                                                                                                    | <i>Aug. 2021 – May 2024</i>  |
| <b>[Completed]</b> "Research on Privacy-Preserving Federated Learning"<br><b>Funded by:</b> National Research Foundation of Korea (NRF)                                                                                                                                                                                 | <i>Jun. 2021 – Feb 2022</i>  |
| <b>[Completed]</b> "Development and Theoretical Study of Security Protocols in Integrated IoD (Internet of Drones) Environments"<br><b>Funded by:</b> National Research Foundation of Korea (NRF)                                                                                                                       | <i>Mar. 2021 – Feb. 2023</i> |
| <b>[Completed]</b> "Research on Privacy Protection of Biometric Information"<br><b>Funded by:</b> National Research Foundation of Korea (NRF)                                                                                                                                                                           | <i>Mar. 2020 – Feb. 2021</i> |

## DOMESTIC PATENTS & SOFTWARE

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**[Patent]** "System and Method for Reputation Management of Payment Channel Hub Based on Blockchain"  
**Registration No.** 10-2730269, Nov. 2024.

**[Patent]** "Device and Method for Generating Cancelable Iris Templates Using Column-Based Slicing"  
**Registration No.** 10-2664795, May 2024.

**[Patent]** "Unmanned On/Offline Multi-Proxy Signature Method Using Drone"  
**Registration No.** 10-2578105, Jul. 2023.

**[Patent]** "Watchtower-based Blockchain Payment Channel Hub for System Liveness"  
**Application No.** 10-2024-0118130.

**[Patent]** "Bitcoin Transaction Tracking with Known Tapscripts in Bitcoin Taproot Upgrade"  
**Application No.** 10-2024-0117036.

**[Patent]** "Secure Leader Election in Blockchain Multi-Party Payment Channel Hub Based on Group Key Transfer Protocol"  
**Application No.** 10-2023-0116492.

**[Patent]** "Secure Cascade Vertical Federated Learning Against Passive Label Inference Attack"  
**Application No.** 10-2023-0116489.

**[Patent]** "Securing a Local Training Dataset Size in Federated Learning"  
**Application No.** 10-2023-0027255.

**[Patent]** "System and Method for Reputation Management of Payment Channel Hub Based on Blockchain"  
**Application No.** 10-2022-0119274.

**[Patent]** "Top-k Biometric Authentication System Using hOPE"  
**Application No.** 10-2022-0022316.

**[Patent]** "Training and Securing Mobility Data Through Blockchain-Based Federated Learning"  
**Application No.** 10-2021-0014268.

**[Patent]** "Cancelable Iris Templates Generation Methodology Using Column-Based Slicing"  
**Application No.** 10-2021-0010156.

**[Patent]** "Wireless Mesh Networking Location-Based Service(LBS) Utilizing CCTV"  
**Application No.** 10-2019-0004817.

**[Software]** "2024 Drone-User Key Exchange Software Graphical User Interface"  
**Registration No.** C-2024-018129.

**[Software]** "2024 Drone-User Key Exchange Software Command-Line Interface"  
**Registration No.** C-2024-018128.

**[Software]** "2023 Authentication and Key Exchange Protocol Implementation Prototype"  
**Registration No.** C-2024-018124.

## SKILLS & LANGUAGES

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### Languages

| Language | Proficiency |
|----------|-------------|
| Korean   | Native      |
| English  | Fluent      |

### Programming Skills

Python (Intermediate), C/C++ (Intermediate), Golang (Intermediate), MySQL (Intermediate)

### Nationality

Republic of Korea / Canada

## REFEREES

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- **Prof. Ik Rae JEONG**  
School of Cybersecurity, Korea University  
irjeong@korea.ac.kr  
**Reference Area:** Ph.D. Supervision & Research
- **Prof. Lyse LANGLOIS**  
Director of the International Observatory on the Societal Impacts of AI (OBVIA), Laval University  
Lyse.Langlois@obvia.ca  
**Reference Area:** International Conferences & Engagement
- **Prof. Kim-Kwang Raymond CHOO**  
Associate Dean for Graduate Studies for the College of AI, Cyber and Computing, and Cloud Tech Endowed Professor, University of Texas at San Antonio; Fulbright Distinguished Scholar, University of Iceland  
raymond.choo@utsa.edu  
**Reference Area:** International Conferences & Engagement
- **Prof. Claudio J. TESSONE**  
Chair of the UZH Blockchain Center, and Professor of Blockchain and Distributed Ledger Technologies, University of Zurich | claudio.tessone@uzh.ch  
**Reference Area:** International Conferences & Engagement; Blockchain Scalability Research Interests

*\*All referees have agreed to be listed.*